MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER PRACTICAL EXAMINATION NOVEMBER 2024

MG1DSCPHY100 – Foundations of Physics

Duration: 2 hrs **Maximum Marks: 35**

Record:5 marks

Attempt the question marked 'X'. Change of question is not allowed :30 marks

1	Measure the dimensions (diameter and length) of a metallic rod using a Screw Gauge and Vernier Calliper. Calculate the volume and surface area of the rod, considering random errors and error propagation.	[A,S][6]
2	Compare the readings obtained using a screw gauge and a vernier calliper by measuring the diameter and thickness of a small cylindrical object (e.g., a thin metal rod or wire). Compare the readings obtained using a vernier calliper and a meter scale by measuring the length and width of a larger rectangular object (e.g., a metal or wooden block).	[A,S][6]
3	Measure the thickness of the given wire using (1) a screw gauge and (2) a microscope.Record the readings taken in both cases with proper calibration. Compare the results obtained and the errors present in the two methods.	[A,S][6]
4	Calculate the mass/density of the unknown liquid by measuring the loss of weight of a submerged object and applying the parallelogram law of vector addition.	[A,S][6]
5	Set up a force table experiment to verify the vector addition of forces. Place the force table at specific angles and attach masses to each string to create forces of known magnitudes.	[A,S][6]
6	Find the height of an object using Laser triangulation method	[A,S][6]
7	Set up a simple pendulum with a length of approximately 1 meter. Using both an analogue stopwatch and a digital stopwatch, measure the time taken for 10 complete oscillations. Record your measurements accurately and calculate the time period T of the pendulum using each stopwatch. Determine the number of significant digits in each measurement and round your results appropriately. Analyze any differences in the time period calculations due to rounding.	[An,S][6]
9	Identify the resistance values of two resistors using the color code. Verify each resistor's value using a multimeter. Then, compare the color-coded tolerance with the measured resistance value.	[An,S][6]

10	Connect the two resistors in series and measure the total resistance using the multimeter. Repeat by connecting them in parallel. Record your observations and calculate the theoretical values for comparison.	[An,S][6]
11	Write a Python program that functions as a basic calculator capable of performing the following operations: addition, subtraction, multiplication, and division.	[An,S][6]
12	Write a Python program to find whether the number given as user input is odd or even / to find the largest number among the three numbers/ to find the sum of the first ten odd numbers	[An,S][6]
13	Write a Python program to print the first twenty number of Fibonacci series/ the multiplication table of 10/ the string "How are you?" using loops.	[An,S][6]